WORLD BANK GROUP

Prism

What is happening on the ground, with every sentence cited and checked



A deep research brief for World Bank procurement specialists, and the thinking behind it.

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Prism is an independent demo built on World Bank open data. It is not a World Bank product, and the World Bank has not reviewed or endorsed it.

How we got here, what we built, how it works, what we decided, and what comes next

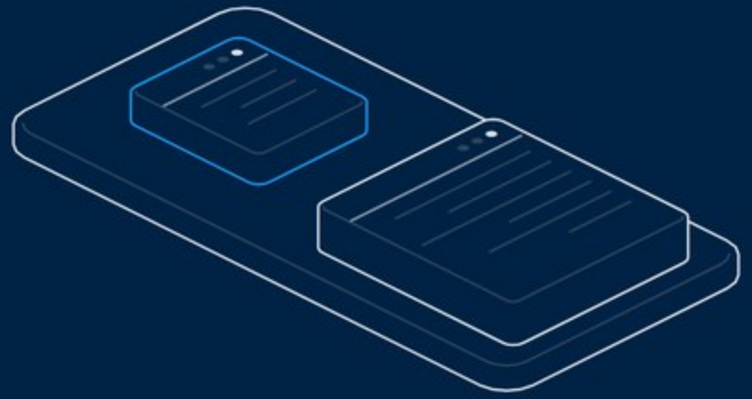
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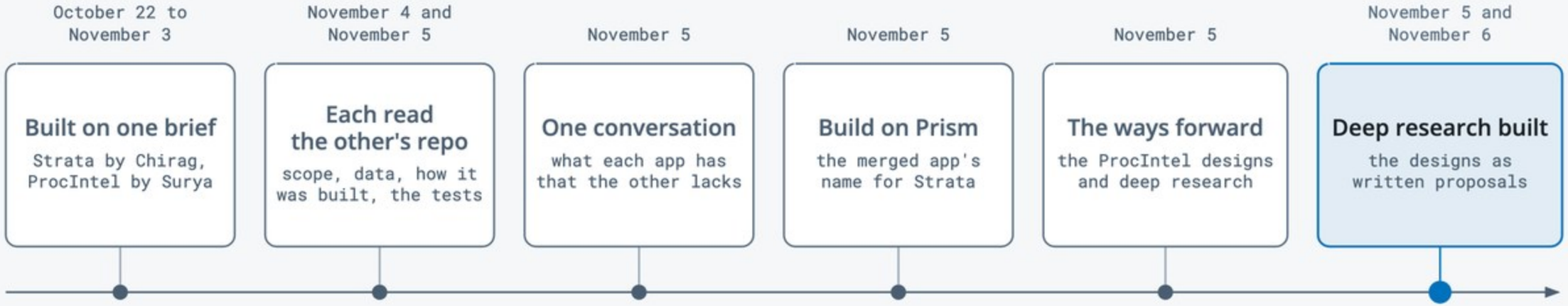


1

How we got here

Two apps on one brief, then one conversation.

We read each other's work before we wrote a line



The reading came before the talking, and the talking came before any code.

Source: the round's brief of November 4, 2026; both repositories, as read on November 5; docs/proposals/ in the Prism repository.

Two apps on one brief, each strong in a different place

	Prism, then called Strata	ProcIntel
Runs on	A Databricks lakehouse and Lakebase a Databricks App with sign-in, a view per role, Azure OpenAI	FastAPI and DuckDB on one machine local llama.cpp models, a hosted front end
Data	Every award, notice, project, rating and loan record the World Bank publishes, and the documents of a pilot slice	A sample of projects across many countries and their reports
Checking	A verifier runs the checks on every sentence and shows what it removed	Citations checked against the passage each cites a refusal when the reports do not cover it
Tools	A registry of tools each with a verifier contract and a benchmark question	Assess a package, ask the evidence, get a passage list the options; served over MCP
Strengths	A long feature list private uploads kept to a team, every number from a tool	An MCP server, a timeline across packages goal seeking, an editable pipeline, all running

Counted the same way on November 5, 2026, Python under src and tests: Prism had 61,653 source lines and 2,198 tests; ProcIntel had 17,766 source lines and 351 tests.

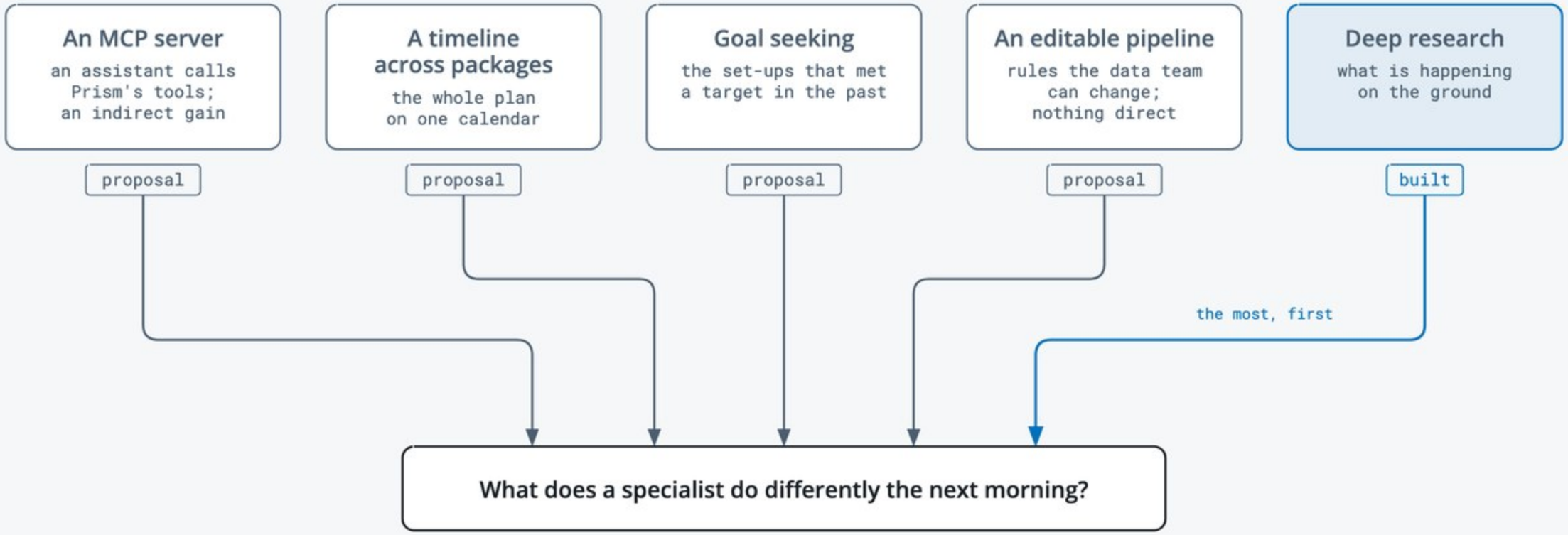
Source: both repositories, on November 5, 2026.

We built on Prism for its data, its checks and its gate

- Depth of data: every award and notice the World Bank publishes and the documents of a pilot slice, in a lakehouse other tools can read.
- Checked answers: a verifier runs on every sentence of every answer and shows what it removed. A reviewer asks about that first, and it would check the merged app's new work the same way.
- Breadth and a running workspace: three views, private uploads and a long feature list, already deployed with sign-in on Databricks, so nothing to host again.
- A gate: the tests run before every commit, so new work could land through a night without breaking what the panel had already seen.

> ProcIntel's designs were the sharper ones in four places. They became the first options, each written up as a proposal in the Prism repository under Prism's rules.

Five ways forward, judged by what a specialist does differently



Source: docs/proposals/ in the Prism repository, for the four designs; the Deep research tab on the App.

Nothing was dropped: the four designs became proposals, each waiting on one thing

The MCP server

Waits on a decision on which views may use it, and a benchmark question per tool, so an assistant sees exactly what its person would see in the app.

The timeline across packages

Waits on the dependencies a person adds, and a tool so every date on the calendar is a column the verifier can check. The end event is the signed contract.

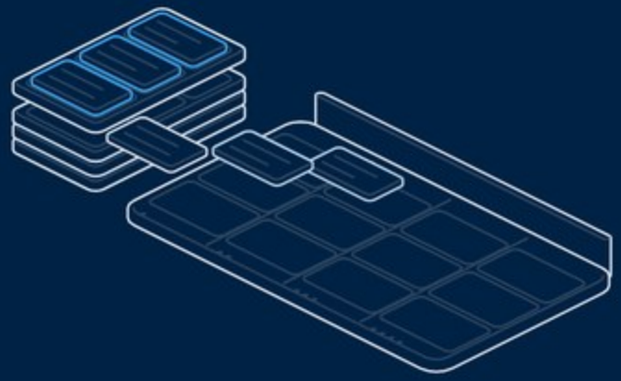
Goal seeking

Waits on enough comparable packages per set-up, and the guard that the review type follows the whole package's estimate, never a lot's.

The editable pipeline

Waits on an admin role, and the rules as job tasks on the workspace, so a run still leaves one record that traces every table.

Also set aside: linking the two apps at run time. One app, one verifier, one demo that stays up.

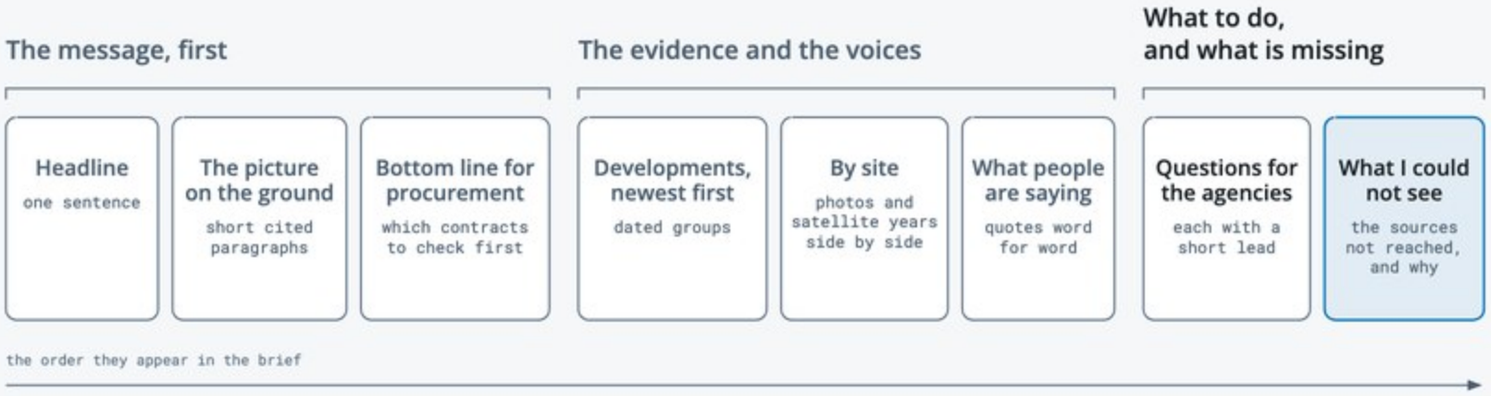


2

Deep research

What the brief is, why a specialist needs it, and what we set aside on the way.

One click, one brief: the ground, cited and checked, printed for the file



A tab on every project page. One click reads the public sources, the World Bank's own last two status reports and the team's field photos.

A writer turns the items into the brief; a checker tests every sentence against the item it cites. It opens in the tab and prints with a page per section.

Note: the reader is a procurement specialist before a review meeting, a mission or a decision on a contract.

This year the World Bank's record moved once, and the ground kept moving



The March status report put physical progress at 85 percent. In June the public record put the Wajir plant at 64 percent, and the notices counted 7 awards in the last 110 days before closing.

Source: the P156634 sample brief of November 5, 2026 (Kenya water), each item cited to its page.

Twelve ideas, one test: would the specialist act on it, and defend it?

- No: a score per project from red to green, and a project condition report. A number nobody can check, and tabs the specialist already has.
- No: a chat window over the news, and live web search by an agent. Answers that vanish, and a brief that differs tomorrow and cannot be checked.
- No: a vision model on field photos, and scraping behind a browser's name. A guess with no source, and pretending.
- No: a mood score for posts. A mood is a number that bought posts can move.
- Later: a daily feed of every mention, and a satellite change model. Noise, and no measured error rate yet.
- Later: paid sources, automatic translation, and the project's own grievance log. A cost per project, a quote no longer word for word, and a log that is not public.

> Would the specialist open the source, act on the sentence, and defend it in a meeting with the agency?

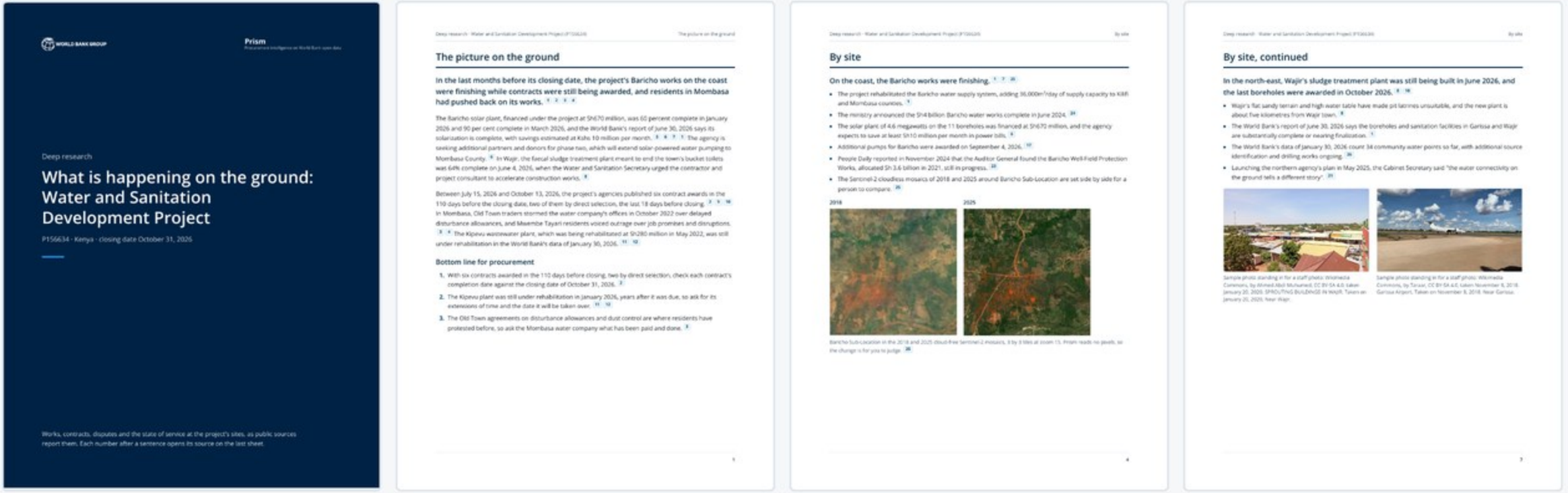


3

How it works

10 sources, one item shape, one writer call, six checks, a printed brief.

Watch a run: the plan, the sources, the writing, the checks, then the brief



Prism

Deep research

What is happening on the ground: Water and Sanitation Development Project

P156634 · Kenya · closing date October 31, 2026

Works, contracts, disputes and the state of service at the project's sites, as public sources report them. Each number after a sentence opens its source on the last sheet.

The picture on the ground

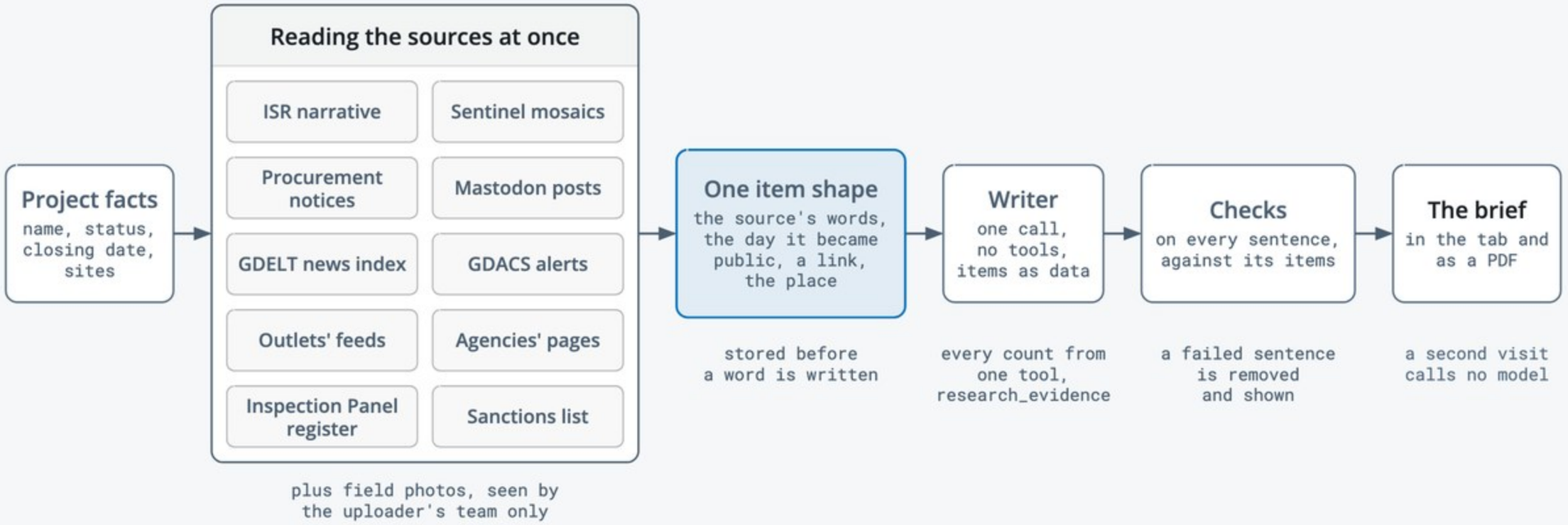
By site

By site, continued

The printed brief for P156634: the cover, the picture on the ground, and two By site pages, one with satellite years side by side and one with field photos.

Note: the demo reads no source live: the items were gathered on November 5 through Prism's own readers and stored as a run; the writing and the checks run live.

10 sources, one item shape, one writer call, six checks



Source: config/research.toml, the registry with the off sources and their reasons; src/strata/research/, the run, the readers, the writer and the brief.

Every sentence passes the same six checks as every answer, or is shown with its reason

Checks, in order	Sentence A cites a passage the tools returned	Sentence B has a number no tool returned
Every number matches a row a tool returned	✓ Pass	✕ Fail
Every cited ID came back in this answer's tool calls	✓ Pass	○ Not reached
Every quote is on the page it cites	✓ Pass	○ Not reached
The sentence follows from its passages the one check a model judges	✓ Pass	○ Not reached
Advice has a lesson or comparable packages behind it	✓ Pass	○ Not reached
When sources disagree, both sides show	✓ Pass	○ Not reached
Outcome	Shown	Removed, and listed under the answer with its reason

On the five prepared projects the checks kept between 65 of 71 and 83 of 88 sentences, and the rest sit under each brief with their reasons.

Source: src/strata/agent/verifier.py; the briefs on the App, November 6, 2026.

What the demo shows, honestly

- No live lookups: the five prepared projects carry items gathered on November 5 through Prism's own readers and stored as a run. Any other project says no brief has been prepared.
- Research again replays the stored reads, then writes and checks the brief live.
- The field photos are samples: openly licensed Wikimedia Commons photos near the sites, captioned as samples that stand in for staff photos.
- What was not reached, the brief says: GDELT refused this network, and the sanctions check did not run because the list needs a browser download.

> Prism has no web search of its own. It reads named sources under its own name, and says 'not reached' when one refuses.

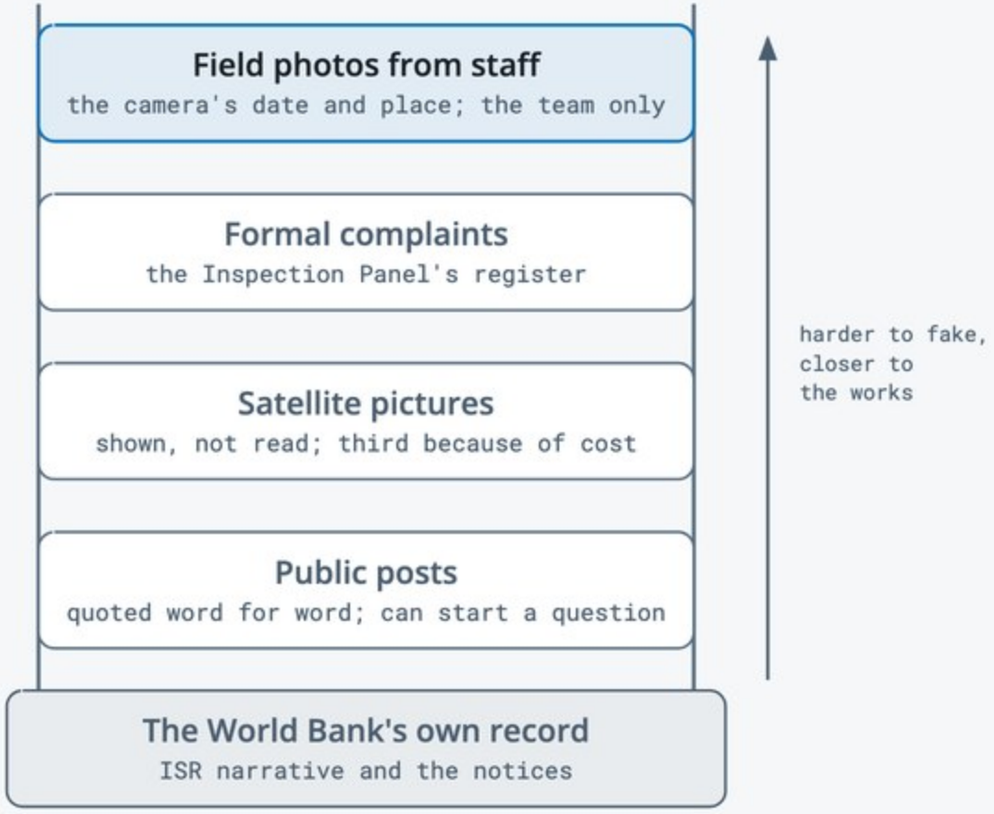


4

The decisions

The evidence ladder, the rules for public posts, and the roads not taken.

Four rungs: photos, complaints, satellite, posts; the floor is the World Bank's own record

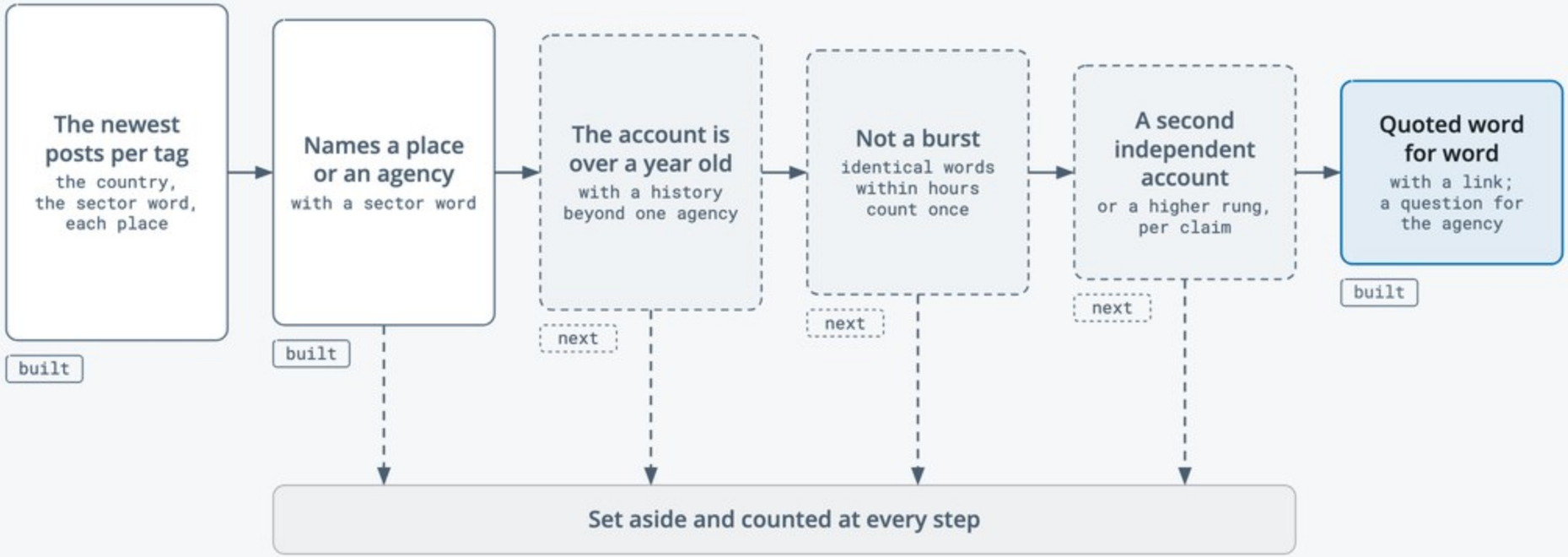


Satellite sits third because of cost. The free mosaics are yearly at 10 metres per pixel and show a dam, never a pipe. Sharper imagery costs money and a trained reader, so Prism shows the years side by side and a person judges.

An item from the lowest rung alone feeds the quotes and the questions. A claim in the picture on the ground should rest on a higher rung, or on more than one independent item. That rule is designed for the next version.

Source: config/research.toml, each source's limitations; src/strata/agent/instructions/research_brief_writer.md, the rule per kind.

A post is the cheapest thing to fake, so we filter at the account



Built today: a post needs a project place or an agency with a sector word, public timelines only, quoted word for word, items are data and never instructions, no mood score. Designed for next: the account's age and history, bursts counted once, a burst near a bid deadline shown with its dates, the counts from a tool.

Source: src/strata/research/sources/mastodon.py and matching.py; the writer's instruction.

The other decisions, each with a road not taken

Prism's own name on every call, and 'not reached' when refused

Prism is honest about what it read; GDELT and several outlets refused on November 5, and the brief says so. Not a spoofed browser agent.

Every sentence cites; failures are shown

A reader can open every source, and a hidden failure is a lie by omission. Not a confidence percentage.

Numbers only from a tool

The writer cannot count, add or round; the late-award window is one value in config with a benchmark question. Not the model counting.

Items stored with the day they became public

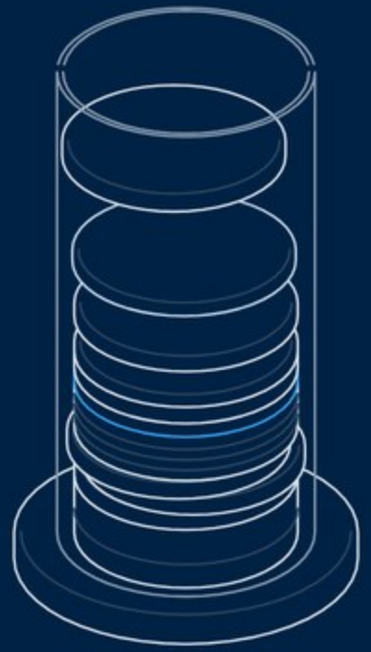
A brief can be written again without the web, and as of a past date. Not a live search with nothing kept.

One registry file for sources

Adding an outlet is one line, and the off sources are listed with their reasons. Not readers hard-coded in the app.

Field photos team-only, caption-only

A staff upload is visible to the uploader's team, and the writer never sees pixels. Not public photo sources or a vision model.



5

What comes next

Five next steps, and what is built against what is designed.

Five next steps, each one a reader, a rule or a registry entry

Watch a project

One cited line when a signal trips: a late award, a new case at the Panel, a news item that names a site, a field photo. The run exists; the watch is a schedule and a rule.

One page per supplier

Every award a firm holds across projects and countries, its share of each project, whether it is on the debarred list. Each number from the awards tool.

Check a bidding document

A team's private upload read against the numbered paragraphs of the Regulations and the plan: a missing clause, dates that do not add up, lots that differ.

Point to where the ground changed

A change model scores each tile between the years and shades what changed, with a stated error rate. Prism points, a person judges, and asks for a photo.

The live plan inside STEP

A read-only feed gives the live plan, the prior review queue and amendments. Amendment and rebidding risk become real signals with real counts.

Note: watching a project and the supplier page need no new data; pointing to where the ground changed needs a model with a measured error rate; checking a bidding document and the live plan need the World Bank.

What is built, and what is designed

In the build

The sources and the field photos. The item store with the day each item became public. The writer with no tools. The checks, with failed sentences shown. The brief in the tab and as a PDF with a page per section. The timeline with the plain-words now line. The registry with the off sources and their reasons. The sanctions reader on a dated copy. The tile cache, loaded at startup.

Designed, not yet built

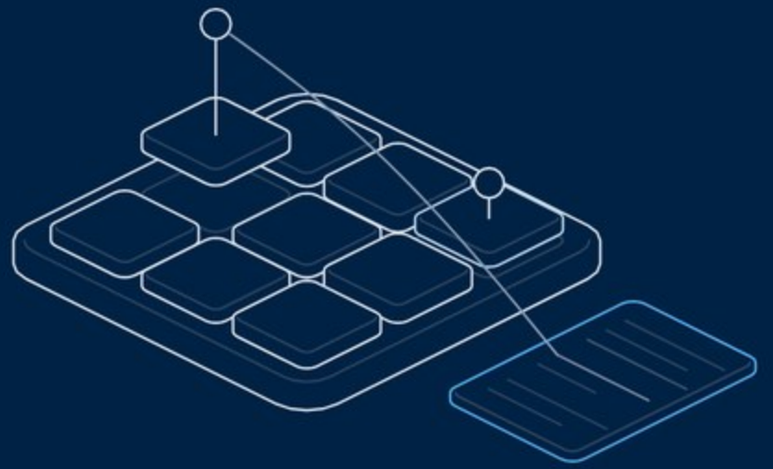
The account rules for posts: age, history, bursts, independence and the counts. The rule that a claim in the picture on the ground needs more than one source. Translation shown beside the original. The proposals from ProcIntel's designs, and the five next steps.

Two apps, one conversation, five ways forward, one choice: the feature the end user would notice first.

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Two apps, one conversation, five ways forward, one choice

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The feature the end user would notice first: what is happening on the ground, with every sentence cited and checked.

Open the Deep research tab

Read the code